Mohammad Hosseini

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EDUCATION

University of Southern California Jan 2023 - Present
PhD in Electrical Engineering; MS in Computer Science *Los Angeles, CA*

- Research: Foundation models and dynamical modeling of neural time series and video data, *NSEIP Lab*
- Advisor: Prof. Maryam M. Shanechi

Sharif University of Technology 2017 - 2022
BSc in Electrical Engineering *Tehran, Iran*

- GPA: 18.81/20; ranked 14th among 180 electrical engineering undergraduates

INTERESTS

Time-Series & Video Modeling
Neural Representation & Decoding
Foundation Models
AI in Health

EXPERIENCE

Graduate Research Assistant Jan 2023 - Present
University of Southern California *Los Angeles, CA*
Advisor: Maryam M. Shanechi, *NSEIP Lab*

- Developed SBIND to disentangle behaviorally relevant dynamics directly from neural images using convolutional recurrent networks and self-attention; outperformed four models on calcium and ultrasound imaging data (ICML 2025).
- Developing preferential models to learn behavior-relevant brain dynamics; implemented bidirectional DPAD and evaluated forward-backward smoothing for neural decoding on Neural Latents Benchmark datasets (NeurIPS 2026).
- Developed WiCAT, a pretrained model for widefield calcium imaging that transfers across subjects, tasks, and datasets and supports zero-shot behavior decoding on unseen subjects; published at ICML 2026.
- Building foundation models for neural time series and video data for brain-machine interfaces, with a focus on data scaling and transfer across subjects and tasks.

Undergraduate Research Assistant Dec 2020 - Feb 2022
BIOSEN Lab, Sharif University of Technology *Tehran, Iran*

- Designed a non-invasive respiratory sensing system with an abdominal force sensor, custom PCB, and Bluetooth Low Energy data acquisition using an nRF microcontroller.
- Designed a stress-induction protocol and collected respiratory data from 12 participants; extracted features and trained classifiers to distinguish stress from relaxation.

Research Intern 2020
Payam-Pardaz Co. *Isfahan, Iran*

- Built a dataset of desktop snapshots and adapted PixelLink for text segmentation as part of a three-person team working on text detection and recognition.

Teaching Assistant 2020 - 2022
Sharif University of Technology *Tehran, Iran*

- Coordinated five laboratory teaching assistants, led discussions for 30 students, and designed and graded assignments across eight courses, including Signals and Systems, Communication Systems, and Computational Intelligence.

PUBLICATIONS

Ongoing Research

- **Foundation models and data scaling for human intracranial recordings.** Hosseini et al. *Preprint; submitted to ICLR 2027.* Building foundation models for neural time series and video data for brain-machine interfaces.

Conference Publications

- Sani, O. G., Jha, T., **Hosseini, M.**, Shanechi, M. M. Preferential dynamic modeling with forward-backward smoothing. ***Advances in Neural Information Processing Systems (NeurIPS)***. (2026), poster. Link.
- **Hosseini, M.**, Erturk, E., Hashemi, S., Shanechi, M. M. Cross-Subject Modeling for Widefield Calcium Imaging via Atlas-Aligned Spatiotemporal Tokenization. ***International Conference on Machine Learning (ICML)***. (2026). Link, Code.
- **Hosseini, M.**, Shanechi, M. M. Dynamical Modeling of Behaviorally Relevant Spatiotemporal Patterns in Neural Imaging Data. ***International Conference on Machine Learning (ICML)***. (2025). Link, Code.

Manuscripts and Preprints

- **Hosseini, M.**, Khatounabadi, H., Fakharzadeh, M. A Low-Power Wearable Respiratory Sensor for Non-Invasive Stress Monitoring. *arXiv:2608.05697.* (2026). Link.

OTHER PROJECTS

Super Marker Tracker (PyTorch): Led a three-person team adapting Facemap for markerless orofacial tracking in mouse behavior videos; introduced an adaptive focal loss that improved keypoint prediction accuracy by 3%.

Multimodal Sentiment Analysis (BERT, ResNet): Combined text and image embeddings through cross-attention and pseudo-label supervision to improve sentiment prediction over single-modality models on the MSCTD dataset.

Neural State Estimation (PyTorch): Implemented Kalman filtering, point-process filtering, and multimodal expectation-maximization methods for neural activity analysis.

TECHNICAL SKILLS

Programming Languages	Python, MATLAB, C/C++, Java, Bash
ML & Deep Learning	PyTorch, TensorFlow, CUDA, BERT, ResNet
Parallel & Distributed Systems	Pthreads, Apache Spark, socket programming
Hardware & Embedded Systems	Altium Designer, nRF52832, STM32, ESP32, BLE

RELATED COURSEWORK

Machine Learning, Deep Learning, Mathematics of High-Dimensional Data, Estimation Theory, Computational Intelligence, Digital Signal Processing, Probability and Statistics, Design of Algorithms, Parallel Programming, Distributed Systems

HONORS & SERVICE

- Awarded the Annenberg Fellowship, University of Southern California.
- Reviewer, International Conference on Learning Representations (ICLR), 2025 and 2026; Neural Information Processing Systems (NeurIPS), 2026.
- Ranked 27th among approximately 150,000 participants in Iran's National University Entrance Exam, Mathematics and Physics track, 2017.
- Member of Iran's National Elites Foundation (admitted in 2017).